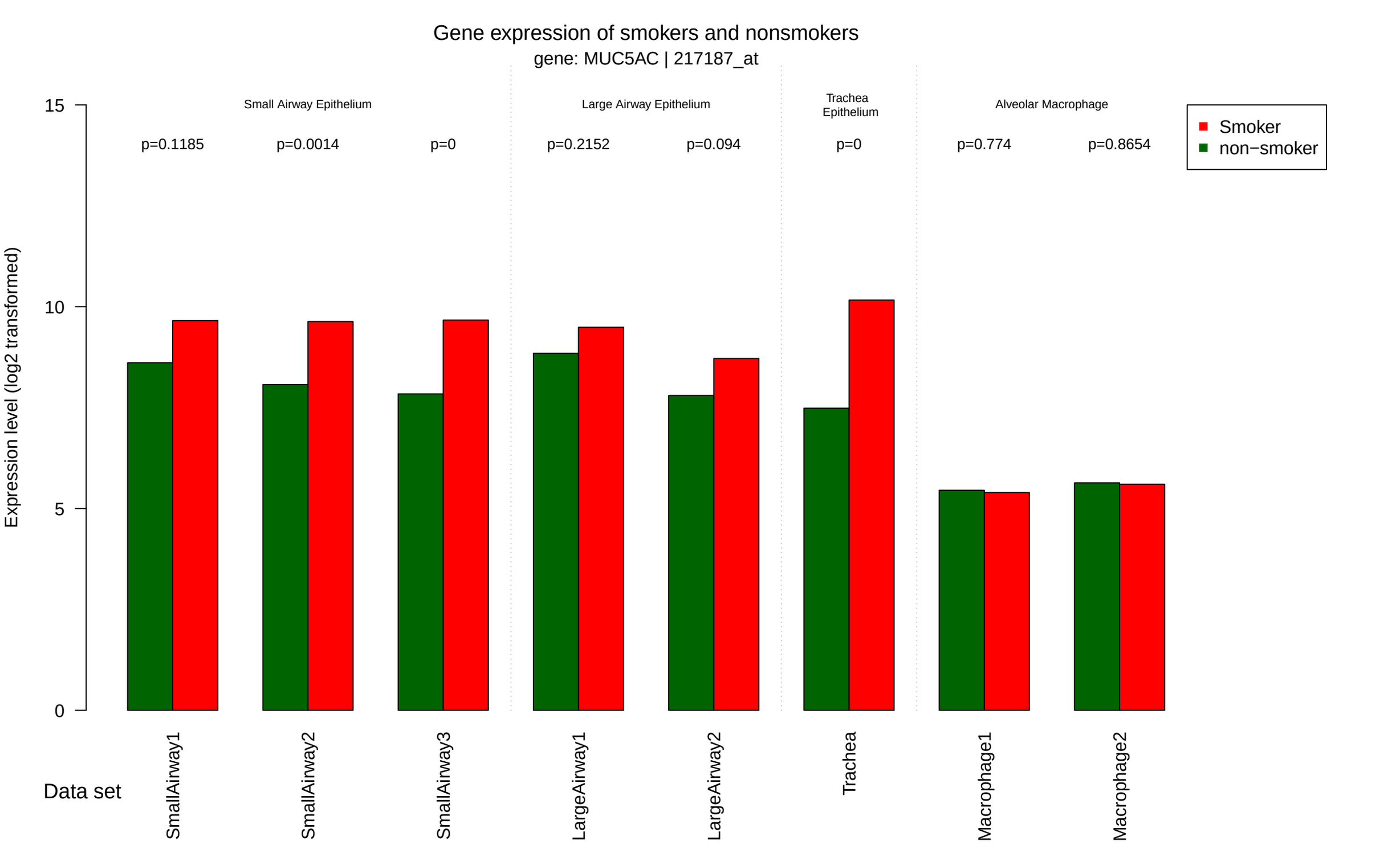
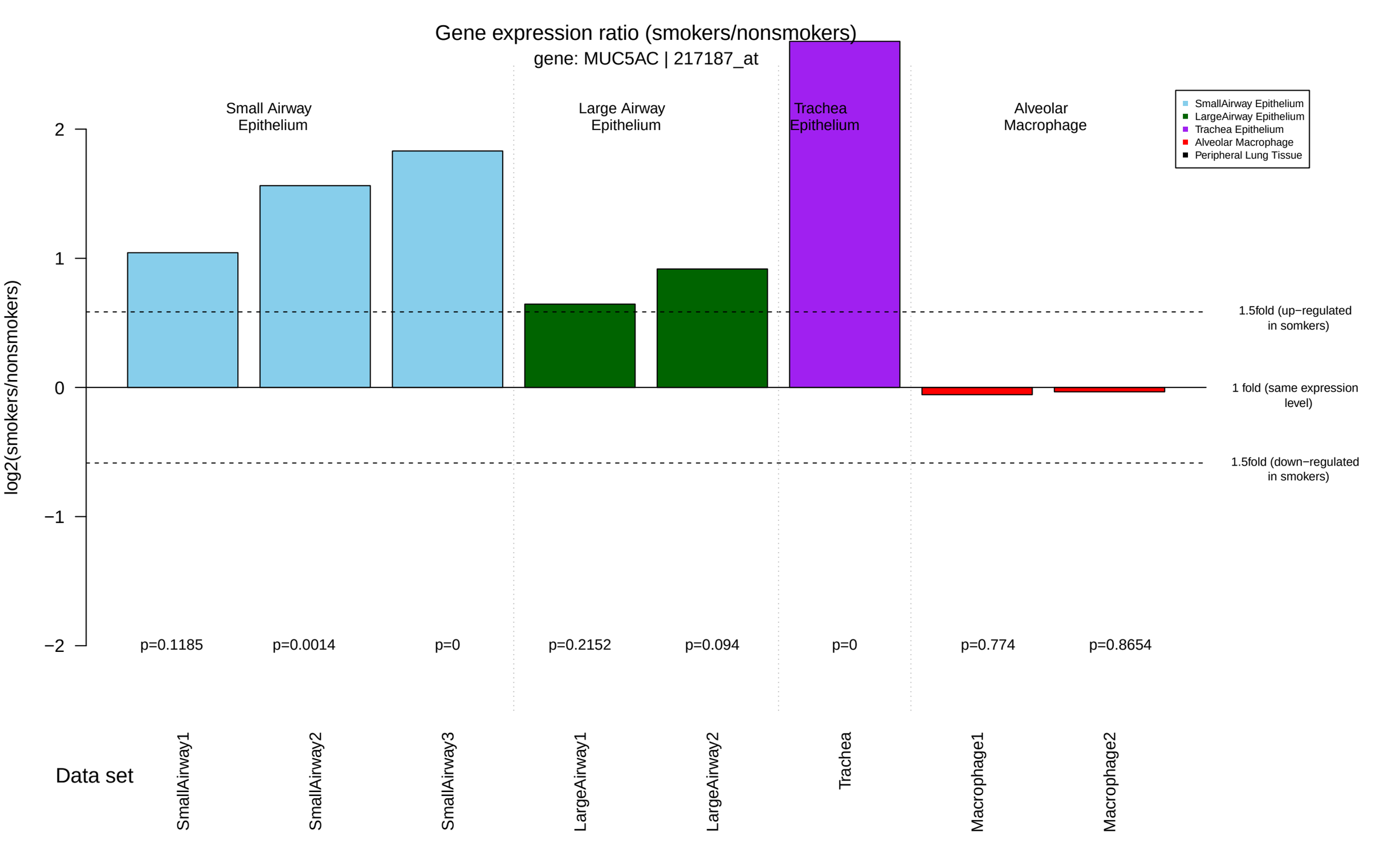
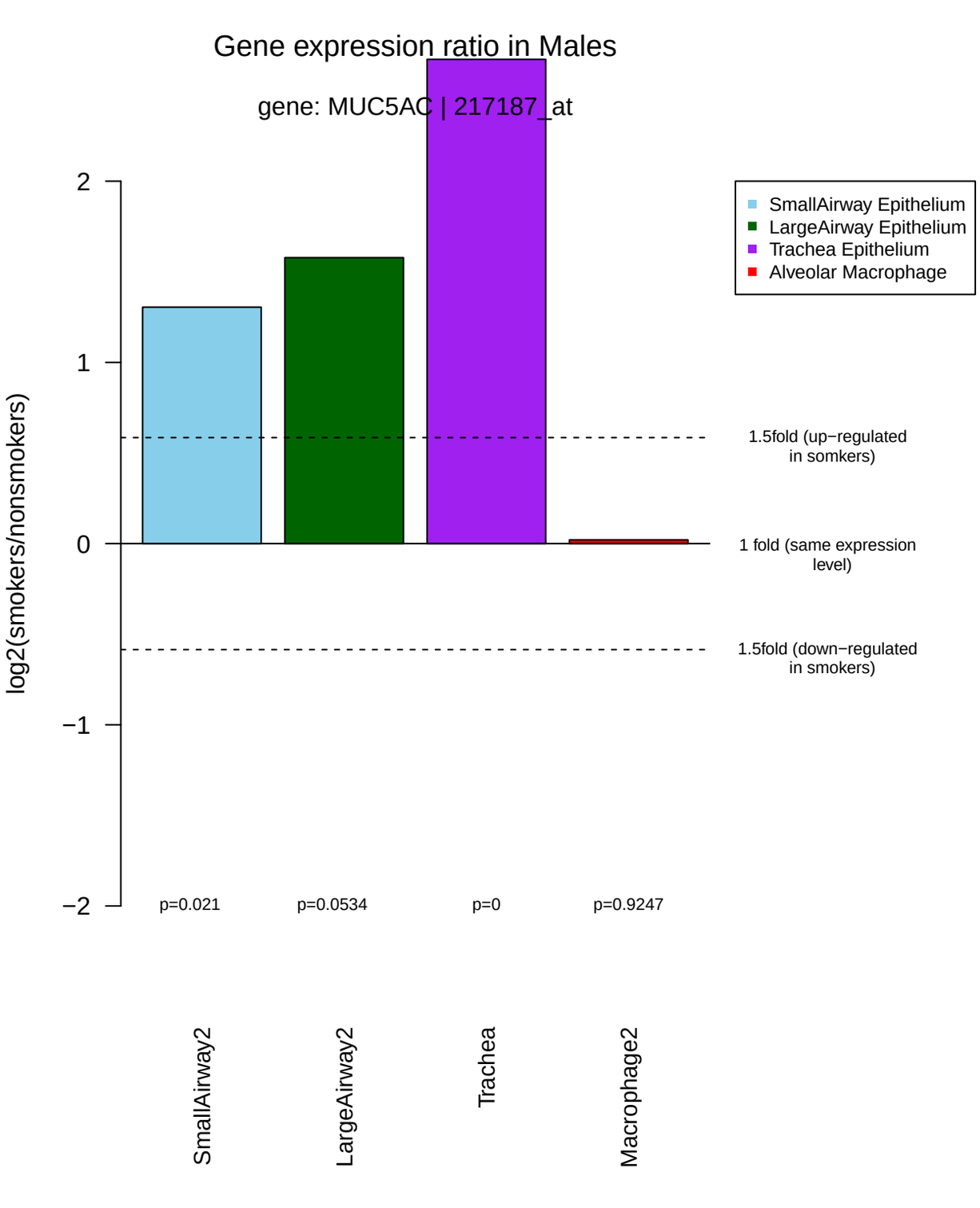
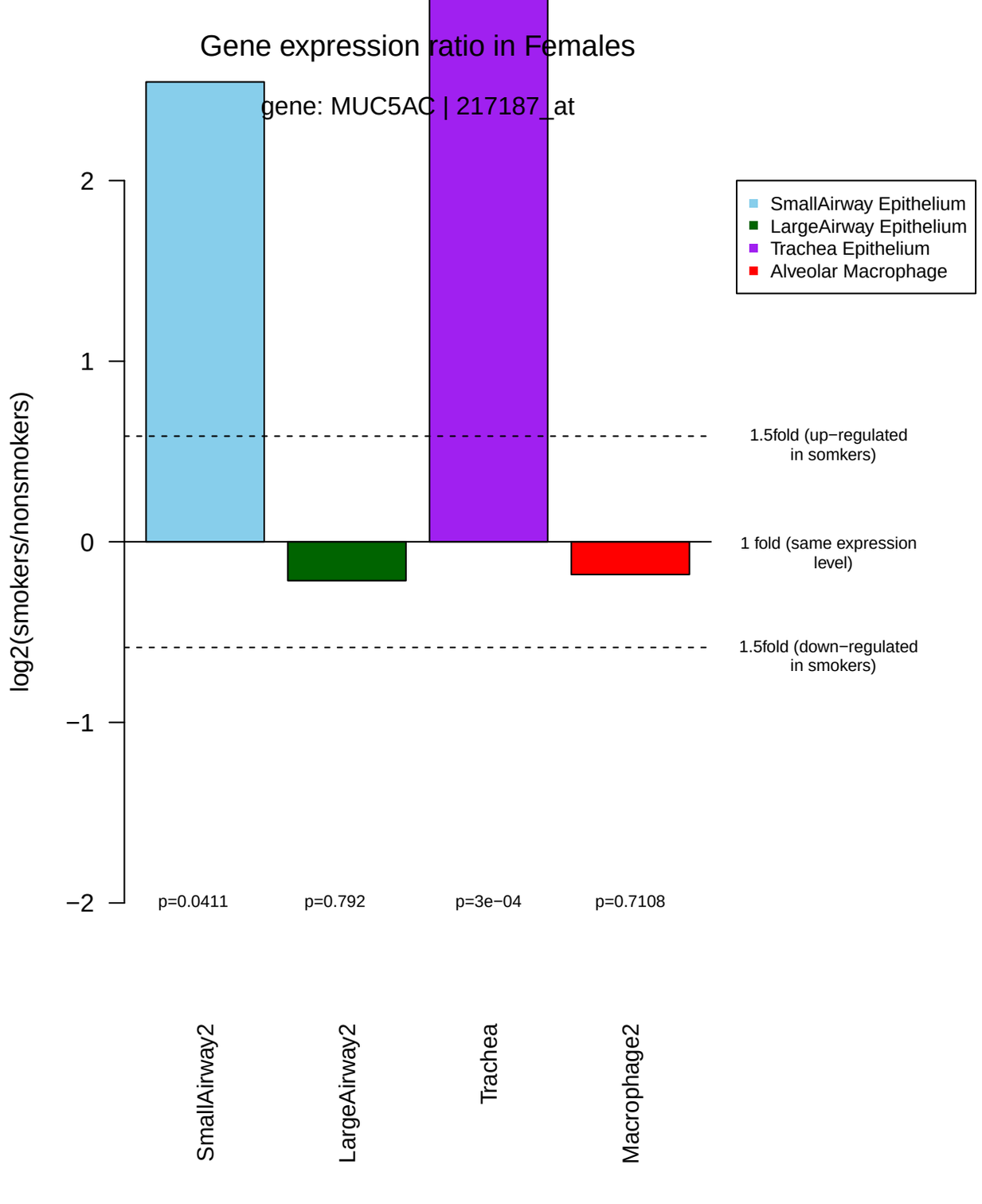
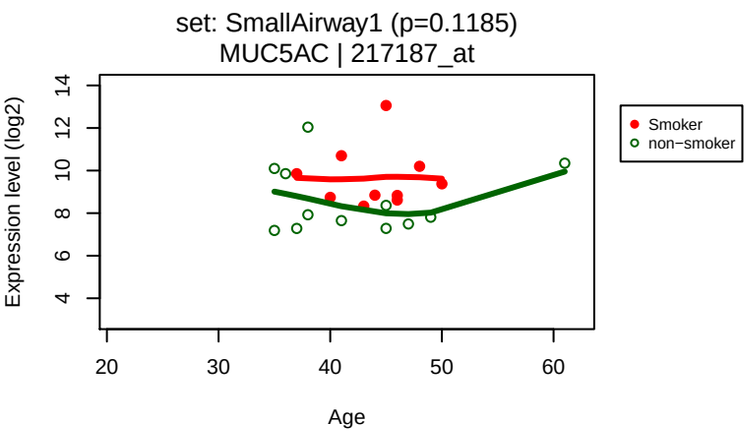
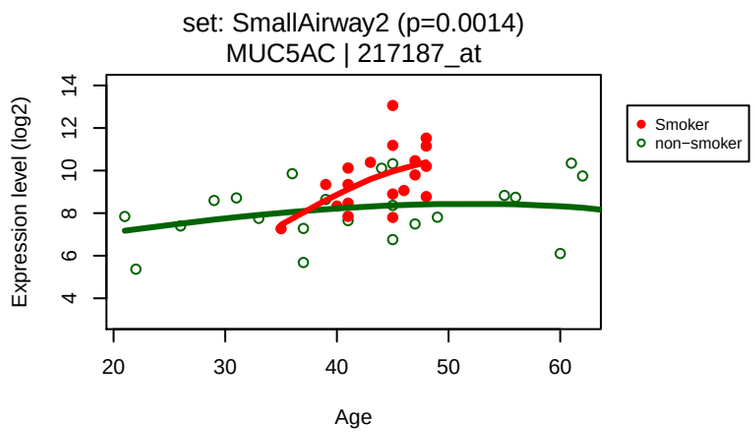

Gene expression of smokers and nonsmokers
gene: MUC5AC | 217187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1185
p=0.0014
p=0
p=0.2152
p=0.094
p=0
p=0.774
p=0.8654
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

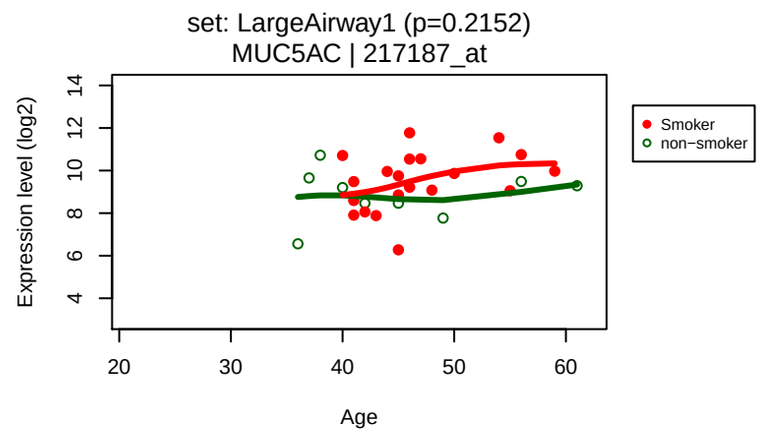

Gene expression ratio (smokers/nonsmokers)
gene: MUC5AC | 217187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1185
p=0.0014
p=0
p=0.2152
p=0.094
p=0
p=0.774
p=0.8654
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

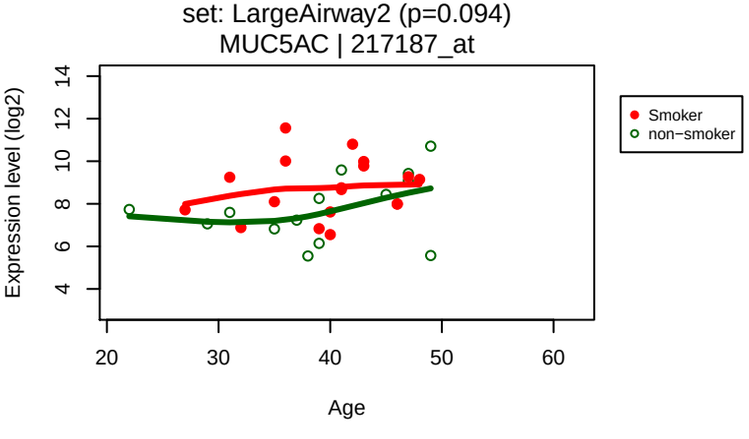

Gene expression ratio in Males
gene: MUC5AC | 217187_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.021
p=0.0534
p=0
p=0.9247
SmallAirway2
LargeAirway2
Trachea
Macrophage2

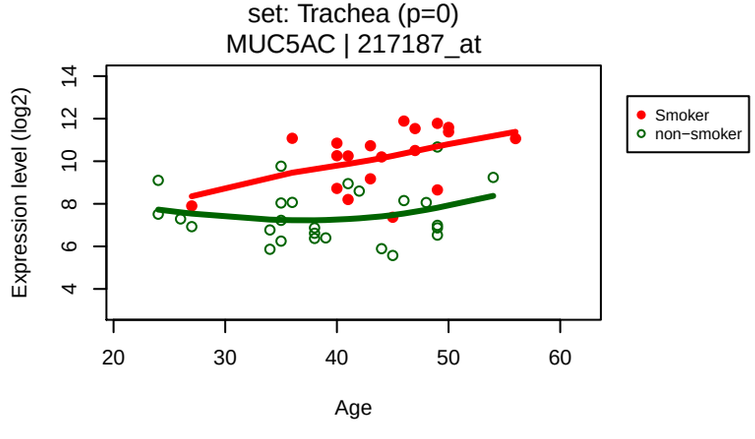

Gene expression ratio in Females
gene: MUC5AC | 217187_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0411
p=0.792
p=3e−04
p=0.7108
SmallAirway2
LargeAirway2
Trachea
Macrophage2

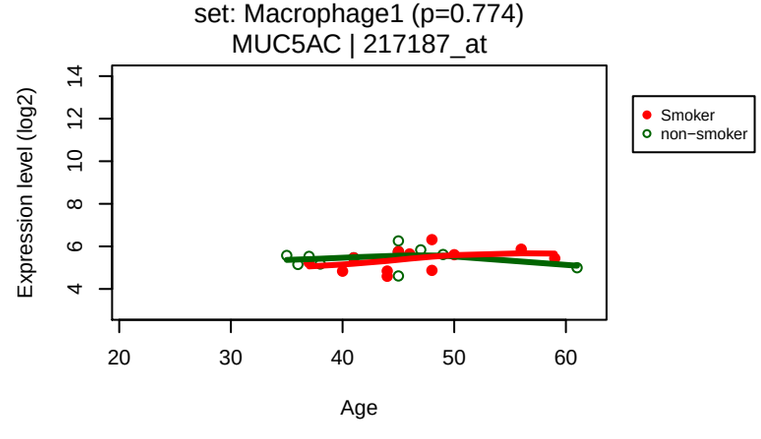

set: SmallAirway1 (p=0.1185)
MUC5AC | 217187_at
Expression level (log2)
Age
Smoker
non−smoker

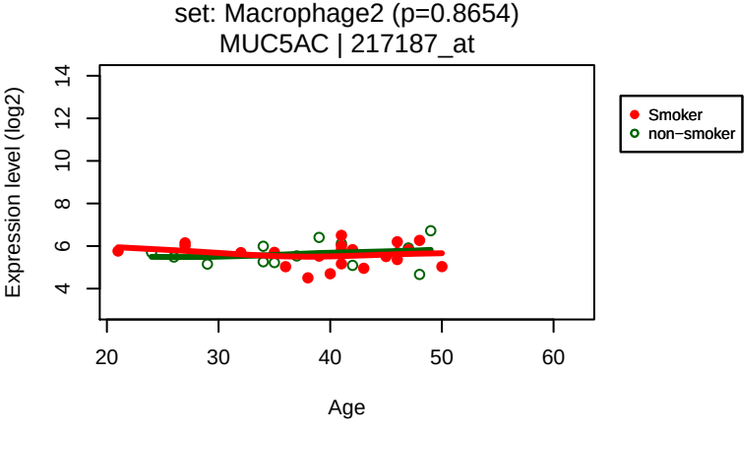

set: SmallAirway2 (p=0.0014)
MUC5AC | 217187_at
Expression level (log2)
Age
Smoker
non−smoker

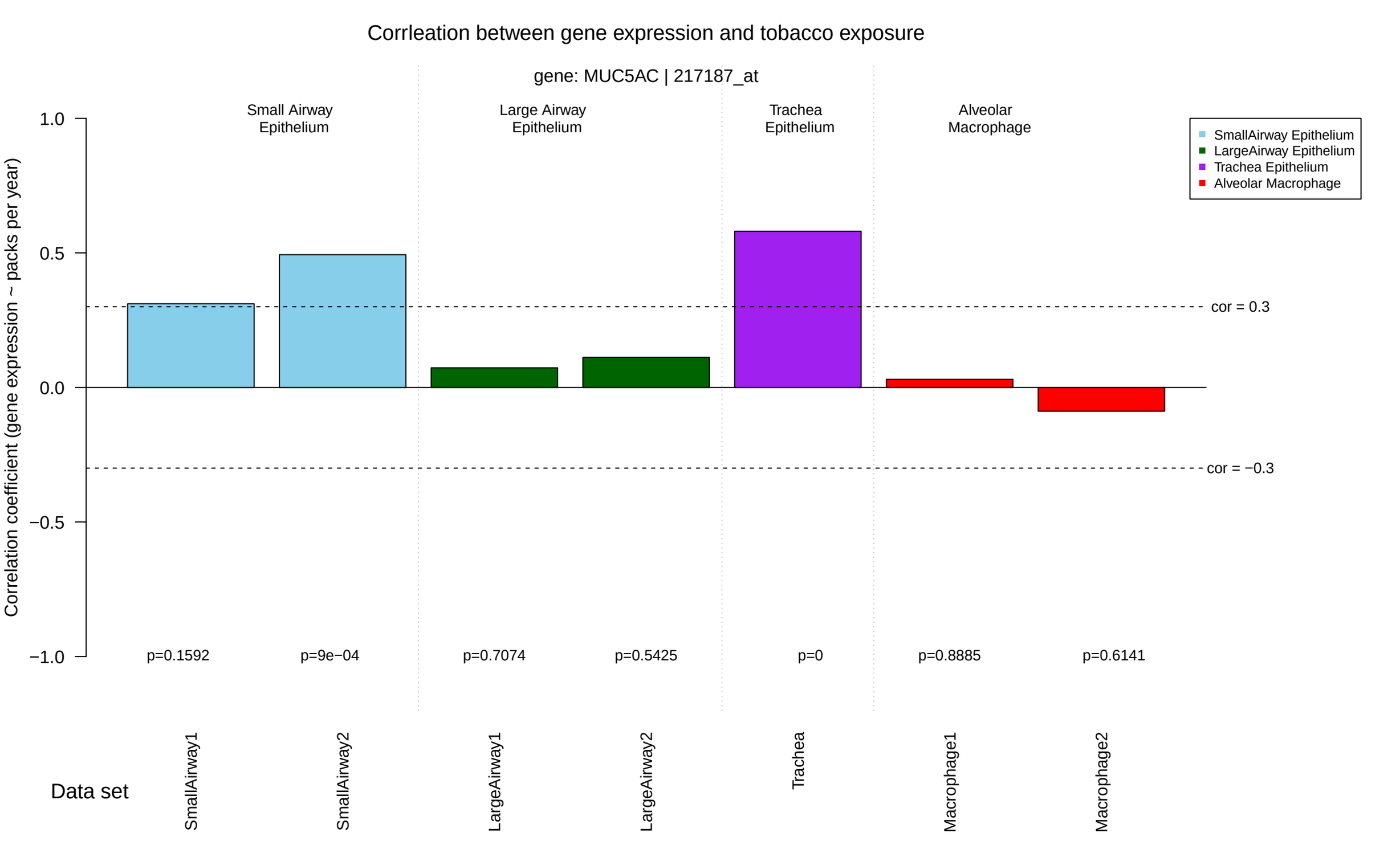

set: LargeAirway1 (p=0.2152)
MUC5AC | 217187_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.094)
MUC5AC | 217187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
MUC5AC | 217187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.774)
MUC5AC | 217187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8654)
MUC5AC | 217187_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC5AC | 217187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1592
p=9e−04
p=0.7074
p=0.5425
p=0
p=0.8885
p=0.6141
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2